

Intelli-Cash VSLA Platform - System Design Document

1. Introduction

1.1 Purpose

This System Design Document (SDD) outlines the architecture, components, and detailed design of the Intelli-Cash platform. It serves as a comprehensive guide for development, implementation, and maintenance, ensuring all stakeholders have a clear understanding of the system's structure and functionality. The document adheres to best practices in software design documentation, drawing inspiration from standards such as IEEE 1016-2009 for Software Design Descriptions [1].

1.2 Scope

The Intelli-Cash platform is designed to digitize and streamline the financial operations of Village Savings and Loan Associations (VSLAs), facilitate their integration into the formal financial ecosystem, and provide robust data analytics for various stakeholders. This document covers the core system components, payment integrations, API ecosystem, data architecture, and security framework.

1.3 Definitions, Acronyms, and Abbreviations

Term	Definition
API	Application Programming Interface
B2C	Business to Customer
C2B	Customer to Business
FLOURISH	Rainforest Alliance FLOURISH VSLA Project
KCB	Kenya Commercial Bank
MFI	Microfinance Institution
NGO	Non-Governmental Organization
PII	Personally Identifiable Information
RBAC	Role-Based Access Control
SACCO	Savings and Credit Cooperative Organization
SDD	System Design Document
VSLA	Village Savings and Loan Association

1.4 References

[1] IEEE 1016-2009 - IEEE Standard for Information Technology—Systems and Software Engineering—Recommended Practice for Software Design Descriptions. Available at: <https://cengproject.cankaya.edu.tr/wp-content/uploads/sites/10/2017/12/SDD-ieee-1016-2009.pdf> [2] Paystack Developer Documentation. Available at: <https://paystack.com/docs/api/> [3] Safaricom Daraja APIs. Available at: <https://developer.safaricom.co.ke/apis> [4] KCB BUNI Developer Portal. Available at: <https://buni.kcbgroup.com/discover-apis>

2. System Overview

2.1 High-Level Description of Intelli-Cash

Intelli-Cash is a digital platform designed to empower Village Savings and Loan Associations (VSLAs) by digitizing their financial activities, enhancing transparency, and providing pathways to formal financial services. It serves as both a group financial management tool and a financial inclusion gateway, connecting VSLAs to external partners such as SACCOs, MFIs, banks, and NGOs. The platform supports the transition from manual record-keeping to full digitization, offering flexibility for parallel use during the adoption phase.

2.2 Key Features and Value Proposition

Intelli-Cash delivers significant value to various stakeholders:

- **VSLAs:** Benefits include simplified record-keeping, increased transparency in financial transactions, and improved access to formal financial services and credit opportunities.
- **Financial Institutions (SACCOs, MFIs, Banks):** The platform reduces lending risk by providing data-driven insights into VSLA performance, savings history, and repayment behavior, enabling more informed lending decisions.
- **NGOs/Programs (e.g., Rainforest Alliance FLOURISH Project):** Intelli-Cash offers real-time monitoring of group performance, impact measurement, and transparent reporting to stakeholders, facilitating data-driven mentorship and evaluation.
- **Ecosystem Partners:** An open API ecosystem fosters innovation and seamless integration with third-party providers, promoting scalable ecosystem growth.

2.3 Stakeholders

The primary stakeholders for the Intelli-Cash platform include:

- VSLA Members
- VSLA Leaders
- Field Officers (supporting VSLAs)

- Financial Institutions (Banks, SACCOs, MFIs)
- NGOs and Development Partners (e.g., Rainforest Alliance)
- Government Systems
- System Administrators

3. Architectural Design

3.1 Overall System Architecture (Conceptual)

The Intelli-Cash platform adopts a modular, service-oriented architecture to ensure scalability, flexibility, and maintainability. The system is conceptually divided into several layers:

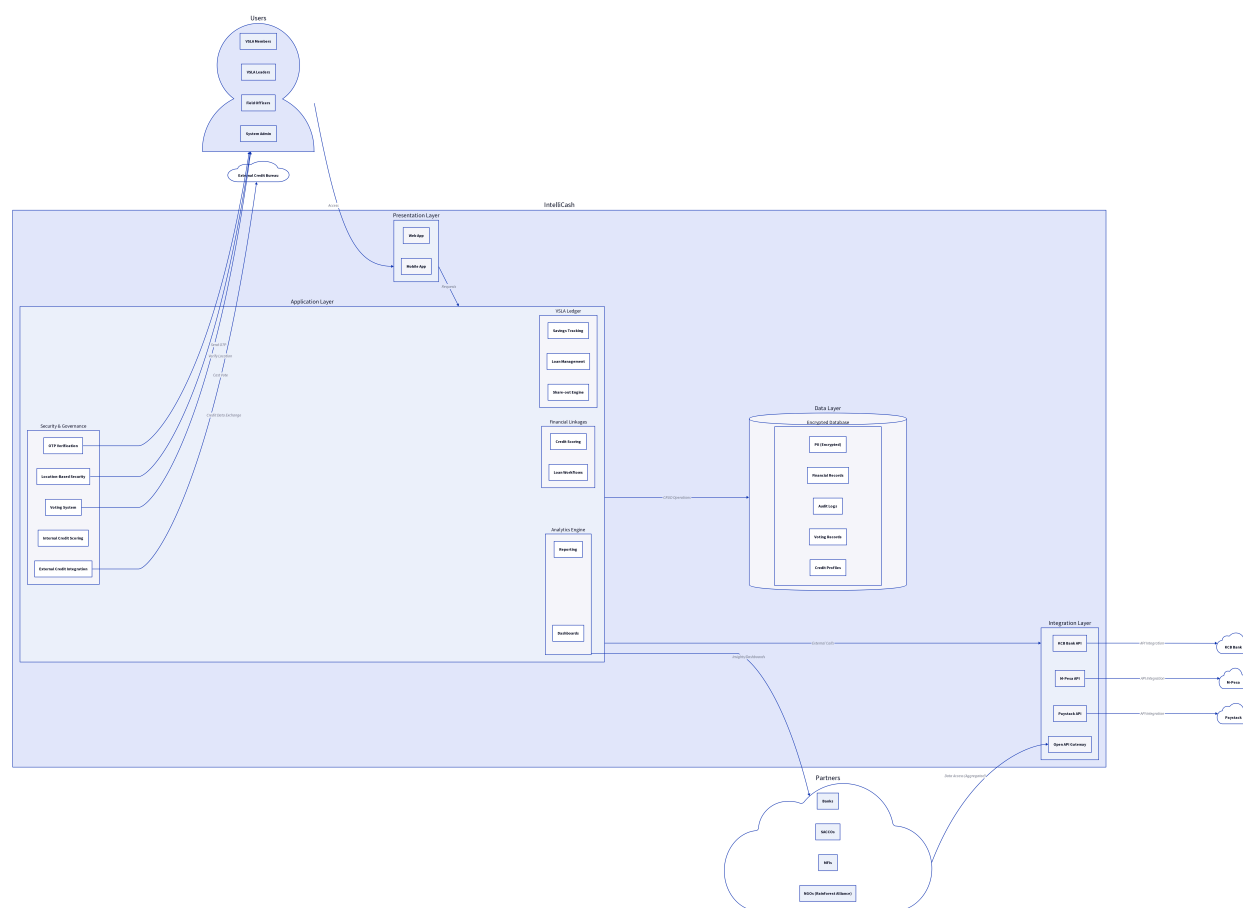


Figure 1: Intelli-Cash System Architecture Overview (Updated)

- 1. Presentation Layer:** User interfaces for various stakeholders (VSLA members, leaders, field officers, partners, admins) accessible via web and mobile

applications.

2. **Application Layer:** Contains the core business logic, managing VSLA operations, financial linkages, and data processing.
3. **Integration Layer:** Handles communication with external systems, including payment gateways, mobile money providers, and financial institutions via APIs.
4. **Data Layer:** Responsible for secure data storage, management, and retrieval, with a strong emphasis on privacy and compliance.
5. **Analytics & Reporting Layer:** Provides real-time dashboards and generates comprehensive financial and performance reports.

3.2 Core Components

3.2.1 VSLA Digital Ledger

This component provides a comprehensive digital solution for VSLA record-keeping, replacing or complementing manual processes. Key functionalities include:

- **Member Management:** Registration, profiles, and attendance tracking.
- **Savings Contributions:** Recording and tracking individual and group savings.
- **Loan Management:** Issuance, repayment schedules, balance tracking, and automated calculations for interest and penalties.
- **Share-out Calculations:** Automated calculation and distribution of share-outs at the end of a cycle.
- **Meeting Records:** Digital logging of meeting activities and decisions.
- **Automated Financial Reports:** Generation of various financial statements and performance reports.

3.2.2 Loan Access & Financial Linkages Layer

This layer integrates VSLAs into the formal financial ecosystem, facilitating access to credit and other financial services. It enables:

- **Partner Integration:** Financial institutions can offer bulk loans to VSLAs or individual credit products to members.
- **Digital Credit Profiles:** VSLAs build credit profiles based on their savings history, repayment behavior, and portfolio quality, enabling data-driven lending

decisions and reducing risk for financial institutions.

3.2.3 Payment Integration Layer

This component ensures seamless financial transactions through integration with multiple payment channels. It supports real-time transaction updates, automated reconciliation, and a secure wallet architecture.

- **Mobile Money Integration (M-Pesa [3]):**
 - **C2B (Customer to Business):** For member deposits and loan repayments. This involves registering validation and confirmation URLs with M-Pesa.
 - **B2C (Business to Customer):** For bulk disbursements of loans and share-outs to members.
 - **Lipa Na M-Pesa Online (STK Push):** Provides a user-friendly method for members to initiate payments directly from their mobile phones.
 - **Group Wallet Management:** Facilitates the management of VSLA group funds.
- **Payment Gateway Integration (Paystack [2]):**
 - Supports card payments and online transactions.
 - Offers multi-channel payment flexibility.
 - Handles currency conversion and settlement for various supported currencies (e.g., KES, NGN, USD).
- **Bank Integration (KCB Bank [4]):**
 - **Direct Bank-to-Wallet Transfers:** Enables seamless movement of funds between bank accounts and mobile money wallets.
 - **Bulk Disbursements:** Facilitates large-scale payouts to multiple recipients.
 - **Account Reconciliation:** Automates the process of matching transactions between the platform and bank statements.

3.2.4 Open API Ecosystem

Intelli-Cash provides secure, controlled APIs to enable integration with a wide range of external systems, fostering a scalable ecosystem. API capabilities include:

- **VSLA Performance Data Access:** Controlled access to aggregated and anonymized VSLA performance metrics for authorized partners.

- **Loan Application and Approval Workflows:** APIs to facilitate the digital submission and processing of loan applications.
- **Transaction Triggers:** APIs for initiating disbursements, repayments, and other financial transactions.
- **Reporting and Analytics Integration:** Allows external systems to pull data for advanced analytics and reporting.

3.3 Data Architecture

3.3.1 Multi-Tier Dashboard Design

The platform features a multi-tier dashboard with robust Role-Based Access Control (RBAC) to ensure data security and usability for different user types:

User Level	Access Permissions
VSLA Members	View personal transactions and loan status only
VSLA Leaders	Full access to their group’s data, including member savings, loans, and meeting records
Field Officers	Access to performance data for assigned groups, supporting mentorship and monitoring
Partners (e.g., Rainforest Alliance, Banks)	Aggregated and anonymized insights into VSLA performance trends, portfolio health, and gender participation metrics. No individual-level exposure without explicit consent.
System Admin	Full system oversight, configuration, and management capabilities

3.3.2 Privacy-First Data Design

Adhering to data protection standards, including the Kenya Data Protection Act, Intelli-Cash implements a privacy-first approach:

- **Data Encryption:** Personally Identifiable Information (PII) is restricted, encrypted at rest and in transit.
- **Aggregated Metrics:** Partner dashboards primarily display aggregated metrics (e.g., repayment rates, savings volumes) to prevent individual-level data

exposure.

- **Consent-Driven Data Sharing:** Data sharing is strictly governed by user consent, partner agreements, and role-based permissions.

3.3.3 Analytics & Reporting

The system provides real-time analytics and reporting capabilities through dashboards, enabling evidence-based decision-making and program evaluation. Key metrics include:

- Savings growth trends
- Loan portfolio performance (e.g., Portfolio at Risk - PAR, repayment rates)
- Member participation and engagement
- Gender and youth inclusion metrics
- Credit readiness scoring for VSLAs and members

4. Security & Compliance Framework

Intelli-Cash is built with a strong emphasis on security and compliance:

- **End-to-End Data Encryption:** All data, from input to storage and transmission, is encrypted to protect sensitive financial and personal information.
- **Secure Cloud Infrastructure:** The platform is hosted on a secure cloud environment, leveraging industry-standard security measures and best practices.
- **Role-Based Access Controls (RBAC):** Granular access controls ensure that users can only access data and functionalities relevant to their roles.
- **Audit Logs:** Comprehensive audit trails are maintained for all transactions and system activities, providing accountability and facilitating forensic analysis.
- **Regulatory Compliance:** The system is designed to comply with relevant data protection laws, including the Kenya Data Protection Act, and donor data governance requirements.

4.1 Enhanced Security Features

4.1.1 Member Verification and Meeting Security

To enhance meeting integrity and member participation verification, Intelli-Cash will implement a robust multi-factor authentication system:

- **OTP Verification for Meetings:** Before a VSLA meeting commences, an One-Time Password (OTP) will be sent to registered mobile numbers of all participating members. Members must verify this OTP within the platform to confirm their attendance and eligibility to participate in proceedings, including voting.
- **Location-Based Security:** Leveraging GPS data from the KoboToolbox data collection (or similar mechanisms), the system will verify that members are physically present at the registered meeting location. This geofencing capability adds an additional layer of security, preventing remote or unauthorized participation.

4.1.2 Secure Voting System for Leadership Elections

Intelli-Cash will incorporate a transparent and secure digital voting mechanism for the election of VSLA leadership (e.g., Chairperson, Secretary, Treasurer). This system will:

- **Member Eligibility:** Only verified members (via OTP and location) will be eligible to cast votes.
- **Anonymous Voting:** Ensure voter anonymity while maintaining verifiable results.
- **Audit Trail:** All voting activities will be recorded in an immutable audit log for transparency and dispute resolution.

4.2 Credit Rating System

Intelli-Cash will feature an internal credit rating system for VSLAs and individual members, and integrate with external credit bureaus to provide comprehensive financial profiles.

4.2.1 Internal Credit Scoring

An internal algorithm will assess creditworthiness based on:

- **Savings History:** Consistency and volume of savings contributions.

- **Loan Repayment Behavior:** Timeliness and completeness of past loan repayments.
- **Attendance Records:** Regular participation in VSLA meetings.
- **Group Performance:** Overall financial health and discipline of the VSLA group.

4.2.2 External Credit Bureau Integration

To provide a holistic view of creditworthiness and facilitate access to formal financial services, Intelli-Cash will integrate with external credit rating agencies. This integration will allow:

- **Data Sharing (with Consent):** Secure and consent-driven sharing of VSLA and member financial data with credit bureaus.
- **Credit Report Retrieval:** Access to external credit scores and reports for VSLAs and individual members, aiding financial institutions in their lending decisions.

4.3 Governance Enhancements

Beyond leadership elections, the platform will support broader governance functions:

- **Digital By-laws Management:** VSLAs can store and manage their constitution and by-laws digitally, ensuring easy access and version control.
- **Meeting Minute Archiving:** Secure storage and retrieval of meeting minutes, linked to attendance and voting records.
- **Dispute Resolution Support:** Tools to document and track dispute resolution processes within the VSLA.

5. Use Case: FLOURISH VSLA Project

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5. Use Case: FLOURISH VSLA Project

Within the Rainforest Alliance FLOURISH Project, Intelli-Cash supports the digitization of VSLAs, enabling:

- Digitization of 20 VSLAs' financial records and operations.
- Real-time monitoring of group performance and financial health.
- Facilitation of financial linkages with formal institutions.
- Data-driven mentorship and coaching for VSLA groups.
- Transparent reporting to project stakeholders and donors.

This results in stronger, more transparent VSLAs, improved access to finance for rural communities, and a scalable model for digital financial inclusion.

6. Conclusion

Intelli-Cash transcends a mere VSLA digitization tool; it functions as a robust financial infrastructure platform. By leveraging secure technology, integrated payment solutions, and data-driven partnerships, it effectively bridges informal savings groups with the formal financial ecosystem, driving financial inclusion and economic empowerment.